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碩士論文計畫書

MASTER'S THESIS PROPOSAL

隱形帳簿：衡量印尼平台交易與收入差距

**The Invisible Ledger: Measuring the Transaction-Revenue Gap of
Digital Platforms in Indonesia**

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1. Research Background and Motivation

Southeast Asia's digital economy has grown rapidly, but platform scale is not captured by platform revenue alone. A marketplace, ride-hailing, delivery, or e-commerce platform can record the full value of a transaction, calculate a payout, and retain only a commission, fee, advertising charge, or related service income as its own revenue. The difference is ordinary accounting. Yet it raises a useful measurement question: how much participant-facing transaction activity is visible in a platform's records but lies outside the revenue that the platform itself recognizes?

Indonesia provides the clearest setting in which to answer that question. Grab, Tokopedia, and Shopee are major intermediaries in the Indonesian market, but their public disclosures do not provide the same kind of country-level evidence. Grab discloses Indonesia revenue but not Indonesia GTV; Tokopedia discloses an e-commerce segment pair that is closely aligned with Indonesia; and Shopee requires country-market evidence combined with a disclosed monetization rate. The asymmetry is not a reason to hide the data. It is part of the measurement problem that the thesis investigates.

I refer to the difference between transaction value and platform-recognized revenue as the transaction-revenue gap, or the invisible ledger. The word invisible does not mean illegal or unrecorded. The gap can include merchant receipts, driver payouts, inventory costs, refunds, and transactions already reported or taxed through other channels. The policy relevance is narrower: a platform may hold an independent transaction record that can, in principle, assist verification of participant reporting. The thesis does not estimate tax evasion, unpaid tax, profit, value added, or GDP undercount.

The project is therefore a source-led measurement study. Its main contribution is to assemble fragmented public evidence into a transparent Indonesia-focused comparison, document the limits of each case, and examine whether the same relationship appears in historical platform disclosures. ASEAN material remains useful as comparative evidence, but it is no longer treated as part of the main sample.

2. Research Questions and Objectives

The proposal asks four linked questions. The first two establish the measurement and its Indonesia-focused application. The third asks whether the relationship recurs in separately disclosed company series. The fourth sets the conditions under which the approach can be extended beyond Indonesia.

Research question	Purpose
RQ1. How can the transaction-revenue gap be measured from public Indonesia-focused platform disclosures?	Develop a transparent source and derivation protocol.
RQ2. What does the protocol show for Grab, Tokopedia e-commerce, and Shopee in FY2023?	Provide a bounded comparative measurement of three Indonesian platform cases.
RQ3. Does the relationship recur in historical platform disclosures?	Use separate annual and quarterly company evidence to assess persistence.
RQ4. When can the same protocol be extended to other Southeast Asian markets?	Treat ASEAN evidence as comparative replication only when equivalent scope and source rules are available.

The objective is not to infer a national tax base from three cases. It is to establish a credible way of measuring a platform accounting and information boundary, show how the boundary differs across leading platform models, and identify what additional country-level disclosure would be needed for wider application.

3. Literature Review and Research Gap

3.1 Platform economics and the location of value

Multisided-platform theory explains why gross transaction value and platform revenue should not be treated as interchangeable. Platforms balance participation across market sides and often retain only a portion of the value transacted through the ecosystem. Rochet and Tirole (2003), Caillaud and Jullien (2003), Armstrong (2006), and Hagiu and Wright (2015) explain the economic logic of intermediation, network effects, and participant-side value creation. A large transaction flow alongside a comparatively small recognized revenue figure is therefore economically plausible, but it still requires careful measurement.

3.2 Information reporting and tax administration

Public-finance research shows that third-party information can materially improve tax-administration capacity. Kleven et al. (2011), Pomeranz (2015), Naritomi (2019), and Kleven, Kreiner and Saez (2016) show why reporting trails and intermediary records matter for verification and compliance. Platform-reporting rules developed by the OECD and the European Union provide the closest policy analogue: the intermediary may report seller or service-provider activity without treating its own corporate revenue as the participant's tax base.

3.3 Research gap

The literature explains both platform intermediation and the value of third-party information. What remains underdeveloped is a source-auditable method for linking transaction value, recognized revenue, and geographic scope in major Southeast Asian platforms. The empirical difficulty is not simply data scarcity. It is that companies disclose the relevant pieces at different levels: country, segment, group, or external market estimate. This thesis makes those differences visible and uses them to define the appropriate scope of inference.

4. Conceptual Framework

Let TV denote gross transaction value recorded by a platform and R denote revenue recognized by that platform. The transaction-revenue gap is TV minus R . The ecosystem ratio is the gap divided by R . A higher ratio indicates more participant-facing transaction activity relative to each unit of revenue retained by the intermediary. It is useful for comparing the accounting boundary within a documented platform scope, but it is not a measure of participant income, taxable income, or value added.

The unit of analysis is a platform-year measurement case. The three FY2023 cases are selected for their relevance to the Indonesian market and for the different disclosure routes through which the same underlying construct can be observed or derived. The purpose of comparing them is not to draw a statistical sample of Indonesian firms. It is to test whether the proposed measurement can be implemented transparently across distinct platform architectures and to identify where the available evidence is stronger or weaker.

5. Data and Research Design

5.1 Main sample and data sources

The main analysis uses a consistent FY2023 Indonesia-focused sample. Each case requires an Indonesia-specific anchor and a documented route to both transaction value and revenue. The underlying package preserves the company filings, source extracts, calculation inputs, scripts, and results separately so that every number can be traced to its source.

Platform	Transaction value	Revenue	Interpretation of evidence
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Platform	Transaction value	Revenue	Interpretation of evidence
Grab Indonesia	Derived from directly disclosed Indonesia revenue and Grab's FY2023 Group revenue-to-GMV rate.	US\$605m, directly disclosed for Indonesia.	Indonesia revenue is direct; Indonesia GTV is a transparent group-rate proxy.
Tokopedia e-commerce	IDR248.836tn, directly disclosed e-commerce GTV.	IDR6.175tn, directly disclosed third-party net segment revenue.	The strongest paired disclosure; an Indonesia-aligned e-commerce segment rather than an explicit country line.
Shopee Indonesia	US\$53.8bn Indonesia market GMV multiplied by Shopee's 40% share.	Indonesia GMV multiplied by Sea's disclosed 10.0% e-commerce monetization rate.	Country GMV is external market evidence; Sea does not disclose Indonesia revenue.

5.2 Calculation and sensitivity

For each case, the analysis subtracts recognized revenue from transaction value and divides the resulting gap by revenue. The principal calculations are mechanical once the source inputs have been selected. Because Grab's GTV and both Shopee quantities require assumptions, the study will show sensitivity to Grab's monetization rate, Shopee's market share, market size, and monetization rate. It will also report the result excluding each platform in turn. These checks do not turn the cases into a statistical sample; they show whether the descriptive result rests on a single assumption or platform.

5.3 Historical and comparative evidence

The historical analysis is deliberately kept outside the main Indonesia table. It uses filing-based annual ratios for Grab and Shopee and a 47 firm-quarter accounting panel for Grab, GoTo, and Sea. These data provide a separate check on whether the transaction-revenue relationship is persistent in company reporting. ASEAN market and policy materials will be used only as comparative context or as an extension if country-level inputs can satisfy the same source and scope rules.

6. Preliminary Evidence and Feasibility

The following figures show that the proposed measurement can be implemented with a clear record of where the evidence is direct and where it is derived. They are preliminary evidence for the proposal examination and remain subject to the advisor's review of the underlying data package.

Case	TV US\$bn	Revenue US\$bn	Gap US\$bn	Gap / revenue	Evidence
Grab Indonesia	5.381	0.605	4.776	7.895x	Derived GTV; direct country revenue
Tokopedia e-commerce	16.331	0.405	15.926	39.296x	Direct Indonesia-aligned segment pair
Shopee Indonesia	21.520	2.152	19.368	9.000x	External market GMV; derived revenue
Selected-platform summary	43.233	3.162	40.070	12.671x	Not an Indonesia-wide total

The current sensitivity checks leave the qualitative result intact. Varying Grab's assumed rate produces a selected-platform gap between US\$39.17bn and US\$41.42bn; varying Shopee's share from 35% to 45% produces a range of US\$37.65bn to US\$42.49bn. The historical evidence includes eight annual company ratios and 47 accounting observations: 17 Grab quarters, 16 GoTo quarters, and 14 Sea quarters. These extensions do not enlarge the Indonesia main sample, but they provide a separate record of persistence and disclosure variation.

7. Expected Contribution, Limitations, and Work Plan

7.1 Expected contribution

The thesis makes three contributions. First, it develops a reproducible way to measure the difference between platform transaction value and recognized revenue from fragmented public disclosures. Second, it documents the practical disclosure limits that determine whether a country-level platform comparison is possible. Third, it connects this accounting boundary to the literature on third-party information: the relevance lies in the potential for transaction records to support verification, not in an assertion that the full gap is untaxed.

7.2 Limitations

The main limitation is the small, source-heterogeneous Indonesia case set. Grab's GTV is derived, Shopee's Indonesian GMV and revenue are constructed from external market evidence and a disclosed rate, and Tokopedia is a business segment rather than a separately labelled geographic pair. The analysis will therefore report platform-level results before any selected-platform summary, preserve all derivations, and avoid claims about Indonesia as a whole. Historical and ASEAN evidence are supporting modules, not substitutes for a consistent main sample.

7.3 Work plan

Period	Planned work
September 2026	Advisor review of the data package; proposal oral examination; confirmation of final sample rule, title, and requested checks.
October-November 2026	Incorporate committee feedback; complete any requested source recovery, sensitivity analysis, and documentation; revise the thesis manuscript.
Before final degree defense	Finalize the manuscript and presentation under the advisor-approved scope; retain ASEAN or investor modules only where their data conditions are satisfied.

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